

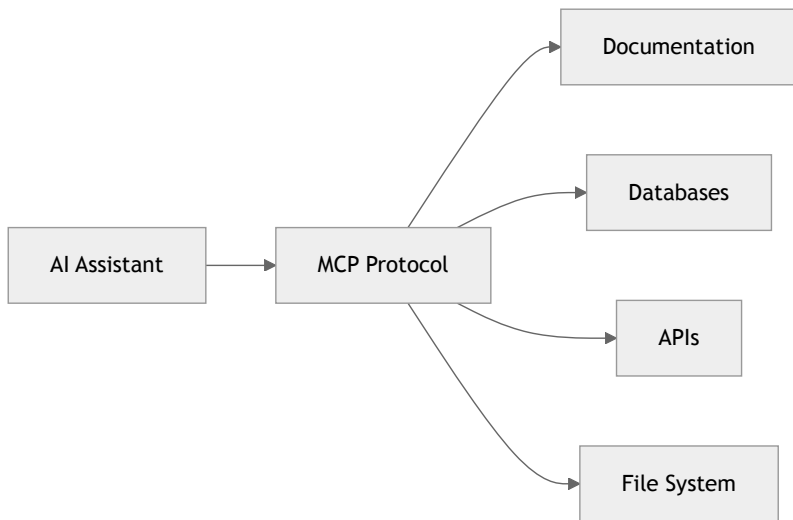
MCP Servers

Connecting AI assistants to external tools and data sources

What is MCP?

Model Context Protocol – an open standard for AI-tool communication

- Universal interface for AI to access external capabilities
- Created by Anthropic, adopted across the industry
- Think of it as **"USB for AI"** – plug in any tool



Why Do We Need MCP?

The Problem: LLMs have a knowledge cutoff date

Without MCP

Outdated code examples

Hallucinated APIs

Answers for old versions

No access to your tools

With MCP

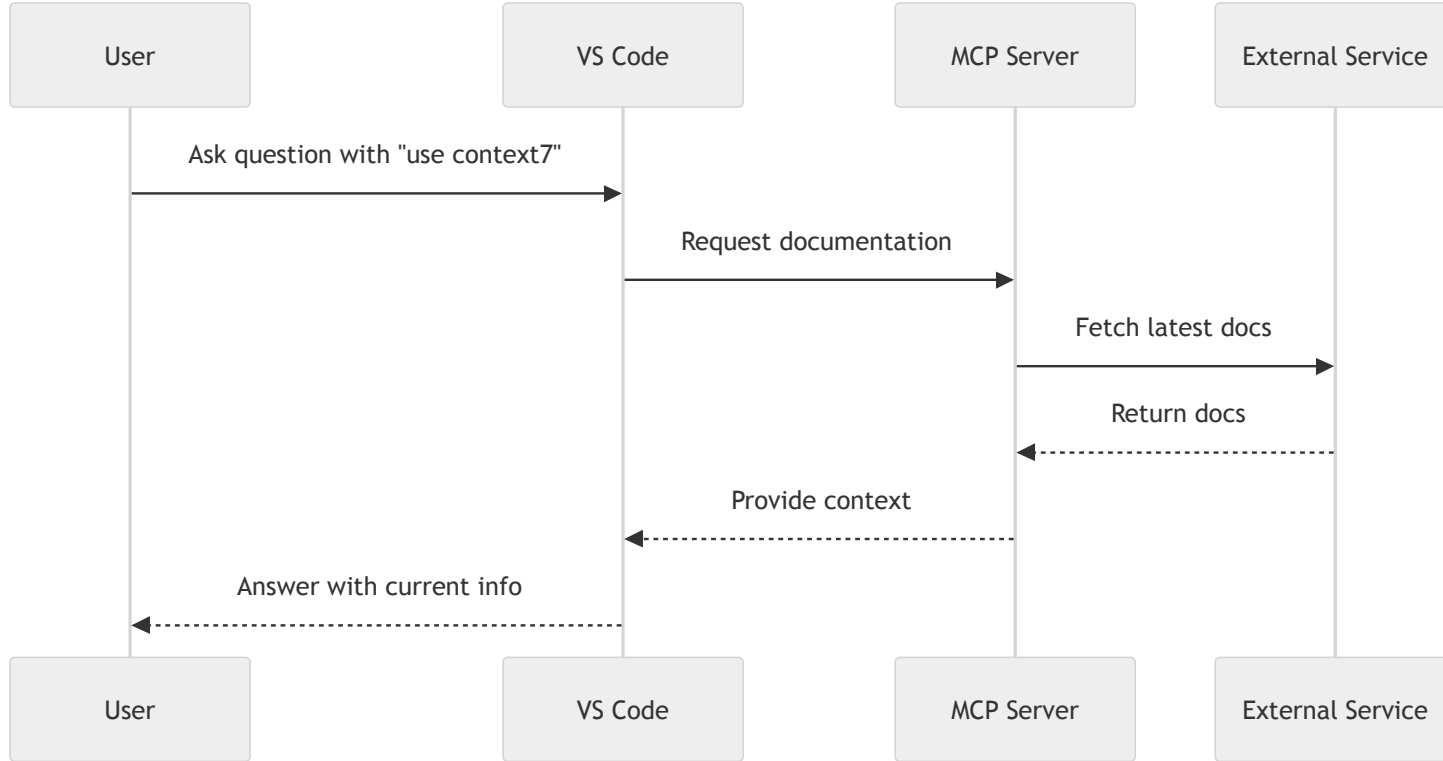
Up-to-date documentation

Real, verified APIs

Version-specific guidance

Direct integration

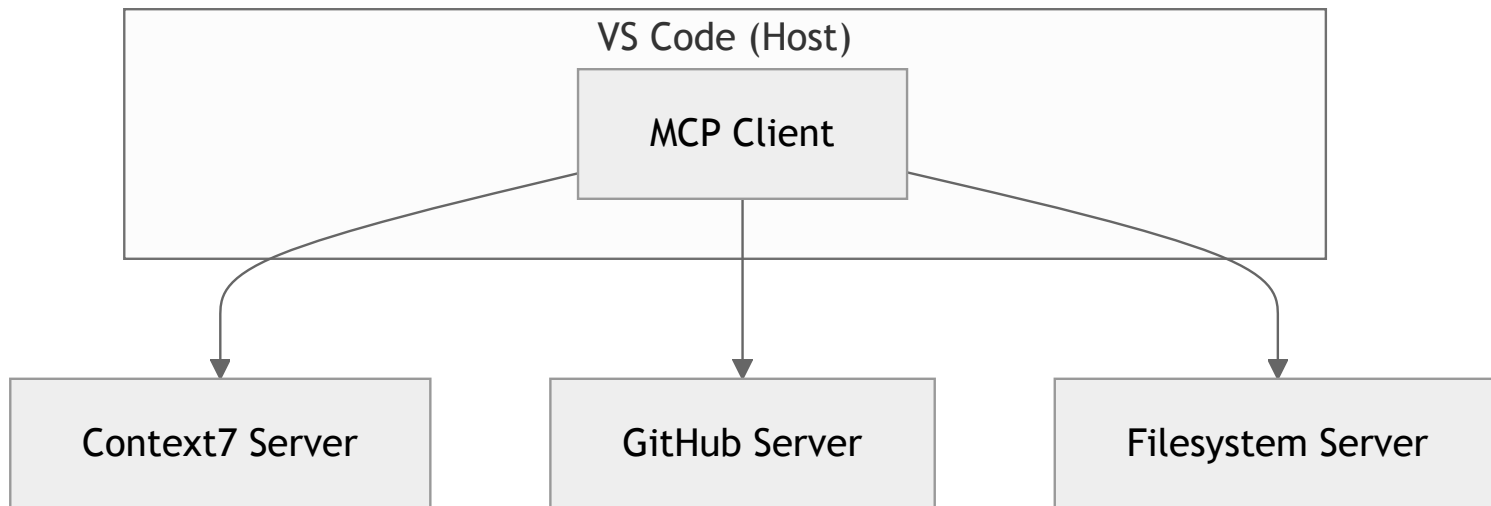
How MCP Works



MCP Architecture

Three key components:

- **Host:** The application (VS Code, Claude Desktop)
- **Client:** Protocol handler inside the host
- **Server:** Exposes tools, resources, and prompts



Examples of Popular MCP Servers

Server	Purpose
Context7	Up-to-date library documentation
GitHub	Issues, PRs, repository access
Filesystem	Read/write local files
PostgreSQL	Database queries
Puppeteer	Browser automation

Browse more: github.com/modelcontextprotocol/servers

What is Context7?

Context7 fetches current documentation directly into your prompts

Available tools:

- `resolve-library-id` – Find the Context7 ID for a library
- `get-library-docs` – Fetch documentation for a specific library

Supports 1000+ libraries including React, Next.js, Express, Prisma, and more

Setting Up Context7 in VS Code

Step 1: Create `.vscode/mcp.json` in your project

```
{
  "servers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp@latest"]
    }
  }
}
```

Requires Node.js >= 18

Starting the MCP Server

Step 2: Start the server in VS Code

1. Open Command Palette (`Cmd+Shift+P`)
2. Run "**MCP: List Servers**"
3. Select **context7**
4. Click "**Start Server**"

Or click the "**Start**" code lens directly in your `mcp.json` file

Verifying the Setup

Step 3: Confirm the server is running

- Open the Chat view (`Cmd+Shift+I`)
- Click the **Tools** icon in the chat box
- You should see Context7 tools listed:
 - `context7 → resolve-library-id`
 - `context7 → get-library-docs`

Getting 403 Errors?

Context7 is rate-limited for anonymous users – a **403** usually means you need an API key.

Symptom	Likely cause
<code>403 Forbidden</code> in MCP output	Anonymous rate limit hit
Tools appear but return no results	Same – unauthenticated
Works once, then stops	Rate limit reached

Fix: get a free API key at context7.com/dashboard, login via your GitHub account and inject it via an env file – **do not hardcode it in** `mcp.json` .

API Key Setup: Using a .env File

Step 1 – Create a gitignored secrets file

Create `.vscode/context7.env` with your real key:

```
# .vscode/context7.env ← never commit this  
CONTEXT7_API_KEY=your_api_key_here
```

Add this line to your `.gitignore` :

```
# Local MCP secrets  
.vscode/context7.env
```

API Key Setup: Using a .env File

Step 2 – Point `mcp.json` at it using `envFile`

```
{
  "servers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp@latest"],
      "envFile": "${workspaceFolder}/.vscode/context7.env"
    }
  }
}
```

Step 3 – Commit a safe example file for teammates

```
# .vscode/context7.env.example ← commit this
CONTEXT7_API_KEY=YOUR_CONTEXT7_API_KEY
```

Restart the MCP server after making these changes

Using Context7 – Basic Usage

Add `use context7` **to your prompt:**

```
How do I set up authentication in an Express API?  
Use context7 to fetch the relevant docs for the version I'm using.
```

Or specify a library directly:

```
Show me how I would create a Prisma schema for my database.  
Use library /prisma/prisma
```

Pro Tip: Auto-Invoke Context7

Add a rule so you don't need to type "use context7" every time

Create `.github/copilot-instructions.md`:

```
Always use Context7 MCP when a request involves a library that the project uses.
```

Hands-On: Try It!

Step 1: Set up Context7

1. Create `.vscode/mcp.json` with the configuration shown
2. Start the MCP server
3. Verify tools appear in Chat view

Step 2: Ask a documentation question

How do I create an Express router in my TypeScript project? use context7

More examples to try:

What's new in React 19? use context7

How do I set up Vitest with TypeScript in a Node project? use context7

Security Considerations

⚠ MCP servers can run arbitrary code on your machine

Best practices:

- Only install servers from **trusted sources**
- Review server configuration before starting
- VS Code prompts for **trust confirmation** on first start
- Check **server output logs** if something seems wrong

MCP: List Servers → Select server → Show Output

Security: What Servers Can Access

Capability	Risk Level	Example
Read files	Medium	Filesystem server
Write files	High	Could modify code
Network requests	Medium	API calls
Execute commands	VERY HIGH	Shell access

Always review what permissions a server needs before starting it.

When NOT to Use MCP

✗ Don't use MCP when:

- Working with **proprietary / internal / confidential** client documentation
- Working in **air-gapped** environments
- **Without express approval** from your project

✓ Use MCP when:

- You need **current** library documentation
- Working with **rapidly changing** APIs or frameworks
- **You have permission** from your project

Troubleshooting

Server won't start?

1. Check Node.js version: `node --version` (need ≥ 18)
2. View server output: MCP: List Servers → Show Output
3. Reset and retry: MCP: Reset Cached Tools

Tools not appearing?

- Ensure server status shows "Running"
- Try restarting VS Code
- Check `mcp.json` syntax

Summary

- ✓ MCP provides a **standard protocol** for AI-tool integration
- ✓ Context7 gives AI access to **up-to-date documentation**
- ✓ Setup is simple: **one JSON file** + start server
- ✓ Always consider **security** when adding MCP servers

Questions?

Resources:

- modelcontextprotocol.io – MCP Documentation
- context7.com – Context7 Website
- [VS Code MCP Docs](#)